

Phenology effects on invasion success: insights from coupling field experiments to coexistence theory

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Abstract. Ecologists have identified a growing number of functional traits that promote invasion. However, whether trait differences between exotic and native species promote invasion success by enhancing niche differences or giving invaders competitive advantages is poorly understood. We explored the mechanisms by which phenology determines invasion success in a California annual plant community by quantifying how the seasonal timing of growth relates to niche differences that stabilize coexistence, and the competitive ability differences that drive dominance and exclusion. We parameterized models of community dynamics from experimentally assembled annual communities in which exotic plants displayed earlier, coincident, or later phenology than native residents. Using recent theoretical advances from the coexistence literature, we found that differences in phenology promote stabilizing niche differences between exotic and native species. However, phenology was more strongly related to competitive ability differences, allowing later invaders to outcompete earlier native competitors and native residents to outcompete earlier invaders in field experiments. Few of these insights could be inferred by comparing the competitive outcomes across invaders, highlighting the need to quantify niche and competitive ability differences when disentangling how species differences drive invasion success.

Key words: annual plants; biological invasions; coexistence theory; fitness differences; niches; phenology; species traits.

INTRODUCTION

Explanations for the establishment and impact of biological invasions often focus on the functional trait and demographic differences between the invader and resident species of the communities they invade. For plant invaders, differences between native and invasive taxa in their maximum photosynthetic rate, specific leaf area, root:shoot ratio, fecundity, relative growth rate, and reproductive effort have all been quantified in efforts to explain invasion success (Daehler 2003, Leishman et al. 2007, Pyšek and Richardson 2007, van Kleunen et al. 2010). However, whether these species differences necessarily advantage the invading species, and when they do, why such advantages arise remains unclear (MacDougall et al. 2009).

The answers to these questions are complicated by the fact that some species differences, termed niche differences, tend to stabilize the interaction between competitors, while other species differences, termed fitness differences, favor competitive exclusion (Chesson 2000, Adler et al. 2007, MacDougall et al. 2009). Moreover, any given trait difference between an invader and resident species will have some stabilizing component, and some component that simply favors or disfavors the

invader. For example, a deep rooting invader may have access to resources unused by the resident community, generating a niche difference. But if more resources are available in deep vs. shallow soils, deeper rooting can also confer a fitness advantage. The relative strength of the niche difference and fitness advantage resulting from a trait difference determines whether that trait difference drives coexistence or exclusion. We note that our use of “fitness” follows from the coexistence literature, where it describes the average competitive ability of a species, and is different from its use in the evolutionary biology literature (Lankau 2011).

The importance of understanding the degree to which niche and fitness differences determine the outcome of invasions is particularly evident in the burgeoning literature asking how invader phenology influences invasion success. The seasonal timing of periodic life-history events such as germination, growth, and reproduction can strongly determine species’ ability to capture resources and convert those resources into offspring. As a consequence, ecologists have often suggested that differences in phenology between the invader and resident species can be an important determinant of invasion success (Bazzaz 1986, Williamson and Fitter 1996, Lloret et al. 2005, Hooper and Dukes 2010, Wolkovich and Cleland 2010). Nonetheless, observational studies comparing the phenology of native and exotic species have found that invaders are sometimes earlier, sometimes later, and sometimes

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coincident in their phenology relative to native residents (Goodwin et al. 1999, Cadotte and Lovett-Doust 2001, Celesti-Grapo et al. 2003, Pyšek et al. 2003, Godoy et al. 2009a, Fridley 2012).

Motivated in part by the ambiguous nature of observational results, Wolkovich and Cleland (2010) argue that the mechanisms by which phenological differences promote invasion are poorly understood, and propose four different hypotheses to clarify this link. Their and prior authors' "vacant niche" hypothesis poses that a unique phenology may allow invaders to establish on resources unused by the resident species (Rathcke and Lacey 1985, Fargione and Tilman 2005), and following Chesson (2000), these correspond to stabilizing niche differences. Although such stabilizing niche differences can favor invader increase, the use of different resources constrains the impact the invader can have on phenologically offset competitors (Hooper and Dukes 2010). Wolkovich and Cleland's (2010) other three hypotheses pose that phenology differences favor the invader over the residents through site preemption, longer periods of activity, or greater flexibility in responding to environmental variation. From a coexistence theory perspective, as presented by Chesson (2000), these are mainly fitness advantages that a phenologically different invader might have over native residents (Gerlach and Rice 2003, Seabloom et al. 2003). Barring other niche differences, these would lead the invader to exclude resident competitors.

In sum, we have reason to expect that phenology differences translate into both stabilizing niche differences and fitness differences in an invasion context. Yet we have little direct experimental evidence to differentiate these alternatives or quantify their contribution to invasion success. Doing so is critical for understanding how not only phenology, but also any trait difference, influences the outcome of invasions (MacDougall et al. 2009). For example, the coexistence of the residents with the invader is predicted when trait differences promote invasion via enhancing niche differences. A very different competitive outcome occurs when trait differences strongly promote fitness advantages; these differences can favor the exclusion of the native community or allow native residents to repel invasion.

Two limitations constrain progress in understanding the mechanisms by which trait differences influence invasion success. First, until recently, ecologists have lacked the toolbox needed to quantify the niche and fitness differences between competitors in the field. Fortunately, recent work from the coexistence literature has shown how these determinants of the outcome of competition can be inferred from mathematical models parameterized with data from experimentally assembled systems of annual plants (Chesson 2000, 2013, Levine and HilleRisLambers 2009). These techniques allow ecologists to quantify the stabilizing niche differences and fitness differences between pairs of invaders and residents differing in their phenological offset. The

second limitation arises from the considerable field effort needed to parameterize these models. As a consequence, these model-based approaches cannot match the species level replication typical of experimental invasion studies or those comparing the traits of native and exotic species. However, the complementary insights provided by quantifying the niche and fitness differences between native and exotic species justify their inclusion among our approaches for understanding the influence of phenology on invasion success.

Here, we couple experiments growing exotic invaders with native communities of differing phenological offset, with mathematical models of annual plant dynamics to dissect how and why phenology differences influence invasion success. Our goal is not to demonstrate the simple importance of phenology in explaining invasion success, but rather to use this trait to demonstrate the complex mechanisms by which trait differences between native and exotic taxa influence the outcome of invasions. We focus on three questions: (1) To what extent do phenological differences between native and exotic species cause stabilizing niche differences vs. fitness differences, and which effect of phenology dominates the outcome of competition? (2) For phenology effects on fitness differences, is an earlier or later phenology optimal for competitive success? (3) To what extent can the influence of phenology on niche and fitness differences be inferred from the outcome of competition alone?

To address these questions, we examined the competitive mechanisms by which phenological differences determine the outcome of invasion in annual plant communities of California. These communities include native taxa and successful exotic invaders, both of which exhibit a wide range of phenology. Additionally, the short and relatively simple life cycle of the annual plants allows us to build models that reasonably describe their population dynamics. We thus quantified the interaction coefficients and vital rates for an early, intermediate, and late-season exotic species competing against three native communities varying from early spring to late summer senescence. These vital rates were used to parameterize a model of competing annual plant species with which the stabilizing niche differences and relative fitness differences could be quantified. We also experimentally varied the relative abundance of each invader within each native community to empirically predict their competitive outcome. We then correlated the niche and fitness differences inferred from parameterized models, the observed competitive outcomes, and the phenological offset between the invader and resident community.

METHODS

Study site and species selection

We conducted our study at the University of California Sedgwick Reserve in northern Santa Barbara County, California, USA (34°40' N, 120°00' W), 730 m

TABLE 1. Species selected for the experiment.

Phenology and status	Family	Species	Seed mass (%)
Early season			
Invasive	Poaceae	<i>Bromus madritensis</i>	
Native	Boraginaceae	<i>Amsinckia menziesii</i>	40
Native	Asteraceae	<i>Chaenactis glabriuscula</i>	20
Native	Asteraceae	<i>Lasthenia californica</i>	40
Middle season			
Invasive	Asteraceae	<i>Centaurea melitensis</i>	
Native	Plantaginaceae	<i>Plantago erecta</i>	33
Native	Lamiaceae	<i>Salvia columbariae</i>	33
Native	Poaceae	<i>Vulpia microstachys</i>	33
Late season			
Invasive	Asteraceae	<i>Lactuca serriola</i>	
Native	Polygonaceae	<i>Chorizanthe palmeri</i>	40
Native	Asteraceae	<i>Hemizonia congesta</i>	20
Native	Polemoniaceae	<i>Navarretia atractylodes</i>	40

Note: Each native community consisted of three different species whose percentage of the total seed mass varied depending on seed availability.

above sea level. The climate is mediterranean with cool, wet winters and hot dry summers. Precipitation was concentrated in winter, and totaled 736 mm over the experimental year (October 2010–September 2011), roughly twice the 50-year average.

Our study focused on three invasive species selected to cover the wide range of phenology exhibited in the system (Table 1). None were dominant at the site of the experimental plots, so their ability to coexist with the focal native communities is uncertain. Like the native grasses and forbs, all three invaders germinate with the first major rains, which occurred in December during the study year, but *Bromus madritensis* (red brome) drops its seeds in May, *Centaurea melitensis* (Maltese star thistle) in July, and *Lactuca serriola* (prickly lettuce) at the end of summer (September). We grew each of these invaders in three different native annual plant communities, each assembled with three species from the reserve selected to have a phenology matching one of the invaders (early, middle, or late; Table 1). We aimed to sow the three native competitors at equal relative abundance (each at 33.33%), but limited seed availability meant that species in the early and late communities were sown at 40%, 40%, and 20% relative abundance (Table 1).

We quantified the timing of aboveground biomass accumulation for all species by sequential harvests of up to eight plants per date over the study period. We calculated the phenological offset between each invader and each native community as the number of days between their dates of peak biomass accumulation. The date of peak biomass for the native communities was the average of the peak biomass dates for their component species. We collected all species' seed within the reserve in spring and summer 2010 from numerous, fully mature, mother plants. We assembled our experimental communities in a 500-m² serpentine soil area that had previously been cleared of all vegetation. Gophers were

excluded for the entire duration of the study, and deer were excluded after their late-season damage to forb species was noted (June onwards).

Quantifying niche and fitness differences between the exotic species and native communities

The assembly of our experimental communities was designed to parameterize a theoretical model. The model is necessary for synthesizing demographic measurements of the plants and their interactions into the stabilizing niche differences and fitness differences that determine the outcome of competition.

The population dynamics of an annual plant with a seed bank can be described (Chesson 1990, Levine and HilleRisLambers 2009)

$$\frac{N_{i,t+1}}{N_{i,t}} = (1 - g_i)s_i + g_iF_i. \quad (1)$$

Here, $N_{i,t}$ is the number of seeds in the soil for species i prior to germination (in the fall or winter). The per capita growth rate for the population ($N_{i,t+1}/N_{i,t}$) is a weighted average of two different growth rates: s_i , the annual survival of ungerminated seed in the soil, and F_i , the viable seeds produced per germinated individual. The weighting term for the average is the germination fraction, g_i . We assume that species affect the performance of one another when germinated individuals limit the fecundity of competitors. Thus the per-germinant fecundity, F_i , can be expanded into a function including the density of competing individuals in the system

$$F_i = \frac{\lambda_i}{1 + \alpha_{ii}g_iN_{i,t} + \alpha_{ij}g_jN_{j,t}}. \quad (2)$$

Specifically, λ_i is the per-germinant fecundity in the absence of competition. It is reduced by germinated individuals of its own and other species, which are multiplied by interaction coefficients, α_{ij} , that describe the per capita effect of species j on species i . The model

ignores the potential for age-dependent survival of ungerminated seeds, because seed bank survival has negligible influence on the competitive outcomes.

Once the model is field parameterized (as described in the next section) we can analyze the per capita growth rate of our invaders when rare (Appendix A: Eq. A.14) in native communities of differing phenology. Moreover, we can follow the techniques of Chesson (2012) to define the stabilizing niche difference, which determines the per capita growth rate advantages species gain when rare, and the average fitness ratio, which determines the degree to which one species is favored over the other in competition. This is detailed in Appendix A, the results of which are summarized here.

Niche overlap, ρ , between any pair of species is

$$\rho = \sqrt{\frac{\alpha_{ij} \alpha_{ji}}{\alpha_{jj} \alpha_{ii}}}. \quad (3)$$

This reflects the average degree to which species limit individuals of their own species relative to competitors, and low niche overlap causes species to have greater per capita growth rates when rare than when common. If species limit individuals of their own species and their competitors equally, then niche overlap is 1, and coexistence is not possible unless species are otherwise identical (Appendix A). At the other extreme, if species have no interspecific effects, then niche overlap is 0. The stabilizing niche difference is $1 - \rho$ and is maximized when species limit individuals of their own species much more than individuals of other species. For example, a winter desert annual would exert much greater limiting effects on individuals of its own species than it would affect a summer annual, and vice versa, generating low niche overlap and a large niche difference.

The average fitness ratio between competitors determines which species wins if there is complete niche overlap ($\rho = 1$). For the annual plant model in Eqs. 1 and 2, the fitness ratio, κ_j/κ_i , is (Appendix A)

$$\frac{\kappa_j}{\kappa_i} = \left(\frac{\eta_j - 1}{\eta_i - 1} \right) \sqrt{\frac{\alpha_{ij} \alpha_{ii}}{\alpha_{ji} \alpha_{jj}}}$$

where

$$\eta_i = \frac{\lambda_i g_i}{1 - (1 - g_i)(s_i)}. \quad (4)$$

If the ratio exceeds 1 then species j wins under perfect niche overlap, if it less than 1, species i wins. Notice that the fitness ratio is composed of two parts, a “demographic ratio” $\eta_j - 1/\eta_i - 1$ and a “competitive response ratio”

$$\sqrt{\frac{\alpha_{ij} \alpha_{ii}}{\alpha_{ji} \alpha_{jj}}}.$$

The demographic ratio describes the degree to which species j produces more seeds ($\lambda_j g_j$) per seed loss due to death or germination ($1 - (1 - g_j)(s_j)$) than species i . The

competitive response ratio describes the degree to which species j is more resistant to competition than species i . Notice that the numerator and denominator differ only in which species is responding to competition, and the arrangement of the interaction coefficients is not the same as in ρ . The demographic ratio and competitive response ratio therefore provide alternative pathways to competitive dominance, and their product (Eq. 4) predicts the winner of competition in the case of perfect niche overlap.

The larger the fitness difference, the greater the stabilizing niche difference necessary for coexistence. To see this, one must first note that species i can invade species j when the product of the niche overlap, ρ and the fitness ratio, κ_j/κ_i is less than one (rearranging the right hand inequality of Eq. A.5 in Appendix A). Given that coexistence requires both species to invade when rare, and that κ_j/κ_i exceeds 1 for one of the competitors, the lesser the niche overlap, ρ , (the greater the niche difference) the easier coexistence is achieved. When there is perfect niche overlap (no stabilizing niche difference), and fitness equivalence, the model collapses to a neutral case (Adler et al. 2007).

Parameterizing the stabilizing niche differences, fitness differences, and per capita population growth rates when rare

Niche and fitness differences determine the outcome of competition between a pair of competitors. In systems with more than two species, these measures can be calculated by treating all competitors faced by a focal species, in our case, the invader, as if they belong to a single species, essentially a community average of the competitors (Levine and HilleRisLambers 2009). Therefore, the competitive effect of each native community on each exotic invader was parameterized irrespective of the identity of the different native competitors in a community. Meanwhile, the vital rates and response of each native community to competition from each invader was an average of the rates and responses of the component native species (as in Levine and HilleRisLambers 2009).

To parameterize our models for each invader competing against each native community, we required estimates of species germination fractions (g_i), per-germinant fecundities in the absence of neighbors (λ_i), seed survival in the soil (s_i), and all pairwise competition coefficients (α_{ij}). In October 2010, we established 124 circular 0.50-m² plots separated by landscape fabric to control weeds. Germination of viable seeds (g_i) was assessed by placing toothpicks adjacent to emerged individuals in repeated censuses of a subset of plots selected to be representative of the entire experiment. These included 10 plots with no competition, and 36 plots with both low (2%, 13%) and high (87%, 98%) invader relative abundance from the competitive outcome plots described in the next section.

We measured seed bank survival (s_i) by estimating seed viability before and after a growing season of burial in 10 nylon mesh bags per species. The bags were buried in November when aboveground seeds began to germinate and be retrieved at the end of summer. To assess viability at these two time points, we placed at least 200 seeds per species on wetted germination paper in a cold room (15°C) for five days, and then kept the seeds at room temperature (22°C) until germination ceased. The remaining seeds were immersed in gibberellic acid for 24 h, and then cut and stained with tetrazolium. Those staining positive were added to the number germinating, yielding a total viability.

To assess the per-germinant fecundities in the absence of neighbors (λ_i) and per capita effect of each invader on itself and on each native community, native community members were grown in the absence of neighbors and with 1.6, 3.2, 7.6, 8.7, and 9.8 g/m² of viable seed of the invader ($n = 1$ plot per competitor density) in 0.5-m² plots. The seeding rates were chosen to give spread in competitor density, and take advantage of the “competitive outcome” plots described in the next section. For the plots with no neighbors, we sowed 100 viable seeds per species, and after germination, we thinned these plots so that each germinant was isolated from competition (>15 cm from the nearest plant). In the competition plots, we identified target germinants that were surrounded exclusively or almost exclusively by the sown exotic competitor. In addition, for estimating the intraspecific invader effects, we selected target invader individuals surrounded by conspecific individuals. For all target individuals (up to eight per plot per species), we measured their seed production (weighing the total mass of seeds and dividing through by the average seed mass), corrected for viability, and then averaged the seed production per species per plot to give an estimate of the per-germinant fecundity, F_i . We then used maximum likelihood methods in R version 2.13.2 (function “optim,” method “L-BFGS-B,” and lognormal error structure [R Development Core Team 2011]) to fit both λ_i and α_{ij} (bounded to be positive) according to the function

$$F_i = \frac{\lambda_i}{1 + \alpha_{ij} g_j N_{j,t}}$$

where g_j was the value determined in other plots. $N_{j,t}$ was the sown density of the competitor species (viable seed number per square meter), though we scaled this density in the fitting process to facilitate convergence. The per-germinant fecundity in the absence of competition, λ_i , should be independent of the identity of the competitor species j , and we therefore fit a common value across the different competitor species.

We used the same overall approach to assess the per capita effect of each native community on itself and each invader. Again, the native community was treated as if it were a single species in competition with the invader, so all fitted parameters, including λ_i , characterize a

“community average” of the species composing a given native community. To ensure that native target species unequally replicated at each competitor density nonetheless contributed comparably to the fitted competition coefficients and λ_i s, we averaged each species’ per-germinant fecundity at a given neighbor density prior to maximum likelihood analyses.

With the empirically estimated germination fractions, per-germinant fecundities in the absence of neighbors, seed survival in the soil, and all pairwise competition coefficients (Appendix B), we calculated the niche differences (Eq. 3) and fitness differences (Eq. 4) between each invader and each native community. In the two cases of “negative” stabilizing niche differences (destabilization), the niche difference term was set to zero (complete niche overlap) for subsequent analyses. For the fitness difference calculations between a given invader and native community, we used the average of the native species’ demographic rates for the community-level rate. These niche and fitness differences were then regressed against the phenological offset (in days) of the invader–native community pairs. We do not present errors for the niche and fitness differences because their component vital rates and interaction coefficients were estimated in different spatial locations, obscuring the meaning of their propagated error.

Empirically estimating the competitive outcomes

To empirically determine the competitive outcomes between each invader–native community pair, we quantified their per capita population growth rates as a function of their relative abundance in sown plots. If the invader or native community has a higher per capita growth rate at all relative abundances, it is the inferred winner. Coexistence is predicted when each grows better at low relative abundance. For each invader–native community pairing, we varied the proportion of viable seed mass sown from 2%, 13%, 24%, 34%, 45%, 55%, 66%, 76%, 87%, to 98% invader. All plots were 0.5 m² and sown with a total of 10 g of viable seed/m², roughly the density annual communities obtain at the study site. Species’ per capita growth rates in each plot (Eq. 1) were estimated with our measures of germination (g), and survival of ungerminated seed (s), as described in the prior section. Per-germinant fecundity, F , was estimated by dividing a species’ viable seed production in a plot by the expected number of germinants (germination rate $\times$ number of seeds added). The native community’s per capita population growth rate was the average of the individual species growth rates.

Inferring competitive outcomes from replacement series designs such as these can depend on the sowing density (Taylor and Aarssen 1989, Levine et al. 2008). We therefore used our theoretical models to corroborate the competitive outcomes. Specifically, we calculated each invader and native community’s per capita population growth rate when rare (Appendix A: Eq. A.12) using the fitted values for g_i , λ_i , s_i , and all

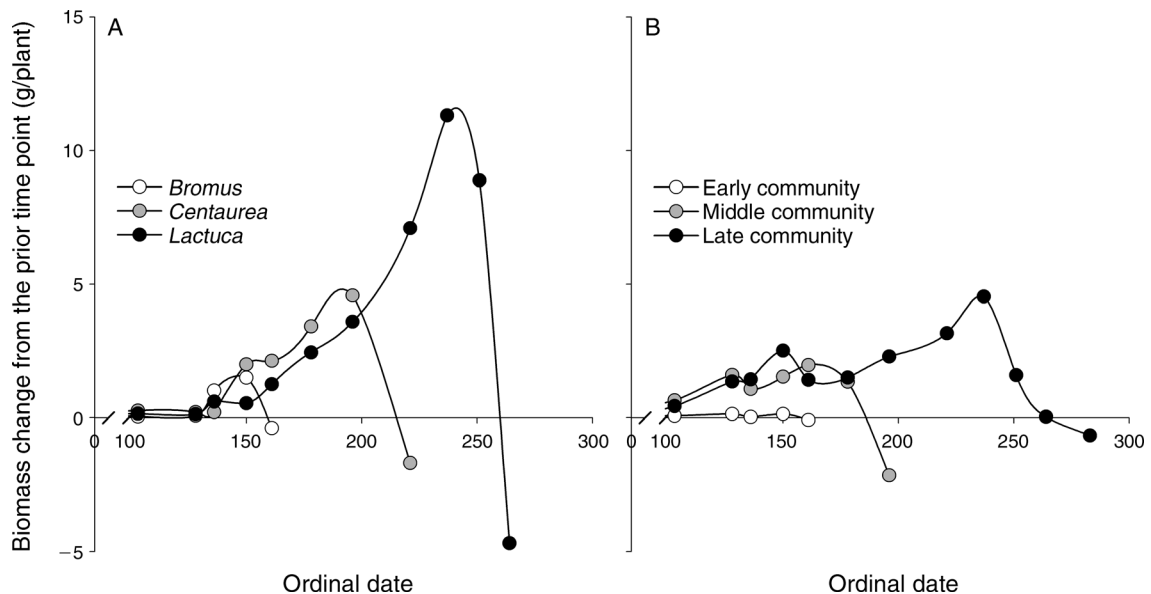


FIG. 1. Changes in aboveground biomass of (A) the invasive species individuals and (B) native community individuals over time. The native community plot shows an average of the per-plant biomass change for the three component species. Because our experimental plots were sown with a fixed seed mass, and species differ in their seed size, these figures are not the plot-level biomass. Open, gray, and black symbols correspond to early, middle, and late phenology.

pairwise competition coefficients (α_{ij}). Coexistence is predicted if both the invader and native community increase when rare; exclusion is predicted if only one has a positive invasion growth rate. For consistency with the empirically determined competitive outcomes, we calculated invasion growth rates assuming an even relative abundance of the three native species per community type.

RESULTS

The three exotic plants differed strongly in the timing of their biomass accumulation, with the early species, *Bromus*, peaking around day 150 (mid May), the middle species, *Centaurea*, around day 200 (early July), and the late species, *Lactuca*, around day 250 (late August; Fig. 1A). The three native communities had biomass accumulation peaks at day 150, 175, and 250, reasonably close to the dates of the three exotic competitors (Fig. 1B). In addition, both later invaders and later native communities tended to produce more biomass per plant than their earlier counterparts. This concordance suggests that the effect of phenology on invader biomass is characteristic of the system.

Invader–native community pairs with greater phenological offset showed both greater stabilizing niche differences (Fig. 2A) and greater average fitness ratios (Fig. 2B) than pairs with less offset. A phenological offset of 100 days, the largest seen between pairs in our experiment, increased the stabilizing niche difference 6-fold as compared to exotic–native pairs with the same phenology (Fig. 2A). We found that the later the invader relative to the native community, the greater the relative

fitness advantage for the invader. In fact, in the three cases where the invader was earlier than the native community, it always possessed an average fitness ratio less than 1, making it an inferior competitor to the native community (Fig. 2B).

The average fitness ratios can be decomposed into a “demographic ratio” and a “competitive response ratio,” whose product is the total fitness ratio (Eq. 4). We found that the demographic ratio between the native and exotic competitors was strongly related to their phenological offset (Fig. 2C), while their competitive response ratio was not (Fig. 2D). Thus later invaders were better competitors, not because they were more resistant to competition, but because they had the capacity to make more seeds per seed loss than their earlier counterparts.

Although the strength of the relationships between phenology and niche differences, and between phenology and fitness ratios were comparable, the fitness ratios increased exponentially with phenological offset (notice the log scale on the y-axis), while the niche differences increased only linearly (Fig. 2A and B). This allows the fitness difference arising from a later phenology to overwhelm the stabilizing niche difference in determining the outcome of competition. For example, a phenological offset of 75 days led to a niche difference of around 0.5 (Fig. 2A). This is strong enough to sustain the coexistence of species with an average fitness ratio of 2 (Appendix A). Yet, results suggests that an invader 75 days later in phenology than the native community would have an average fitness ratio in its favor of

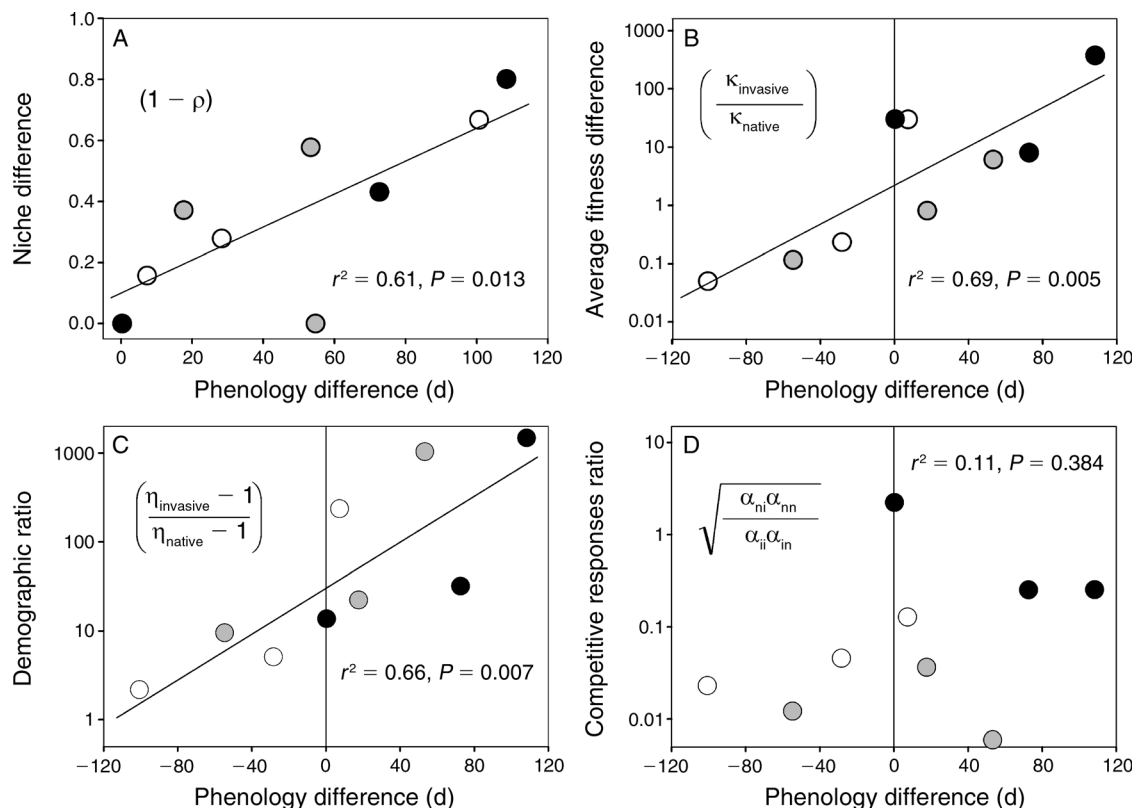


FIG. 2. (A) Niche differences and (B) fitness differences decomposed into (C) demographic differences and (D) competitive response differences between invasive species and the native communities. In panel A, differences in phenology (x -axis) were calculated in absolute value because niche differences should be a function of the trait difference between the competitors, regardless of which species is later vs. earlier. By contrast, fitness differences favor one species over the other. Thus we need to know whether the focal species is earlier or later than its competitor (the sign of the phenology difference) to determine how phenology relates to fitness advantages. Open, gray, and black symbols correspond to pairwise interactions with *Bromus*, *Centaurea*, and *Lactuca*. Note log scale in panels B, C, and D. Variables are ρ , niche overlap; κ , fitness; α_{ij} are interaction coefficients that describe the per capita effect of species j on species i (where n is native and i is invasive); and η is explained in Eq. 4.

around 30 (Fig. 2B), fifteen times greater than what the stabilizing niche difference could sustain for coexistence.

The more rapid rise of fitness ratios than niche differences with phenological offset explains why despite the niche differences (Fig. 2A), our relative abundance manipulation and parameterized models predicted the local scale exclusion of the earlier competitor in the six invader–native community pairs with phenological offset (Fig. 3B–D, F–H). For example, the competitors with the greatest niche difference, the late invader *Lactuca* vs. the early native community, were not predicted to coexist owing to the over 200-fold fitness ratio in *Lactuca*'s favor. By contrast, the one case of predicted coexistence involved a pair with little phenological offset (the middle invader, *Centaurea* and middle native community). Importantly, the theoretically predicted outcome of competition from the parameterized models (Fig. 3 insets) matched the empirically predicted outcome from the relative abundance manipulations for all nine of the competitive pairs (Fig. 3). The high concordance between the theoretically predicted and

empirically determined competitive outcomes affords confidence in our model structure and parameter estimates (Appendix B).

DISCUSSION

Our central finding is that phenology differences between exotic and native species promote invasion success via two distinct pathways. First, they promote stabilizing niche differences. We found that invaders with a 100-day phenological offset from the native community had six times the stabilizing niche difference as those with no offset (Fig. 2A), which gives invaders an advantage when they enter a community at low relative abundance. Moreover, the latest invader showed progressively increasing niche differences from the late to middle to early native community, while the early invader showed progressively increasing niche differences from the early to middle to late native community (Fig. 2A). In fact, only the intermediate-phenology invader deviated from the monotonic increase in niche differentiation with increasing phenological offset.

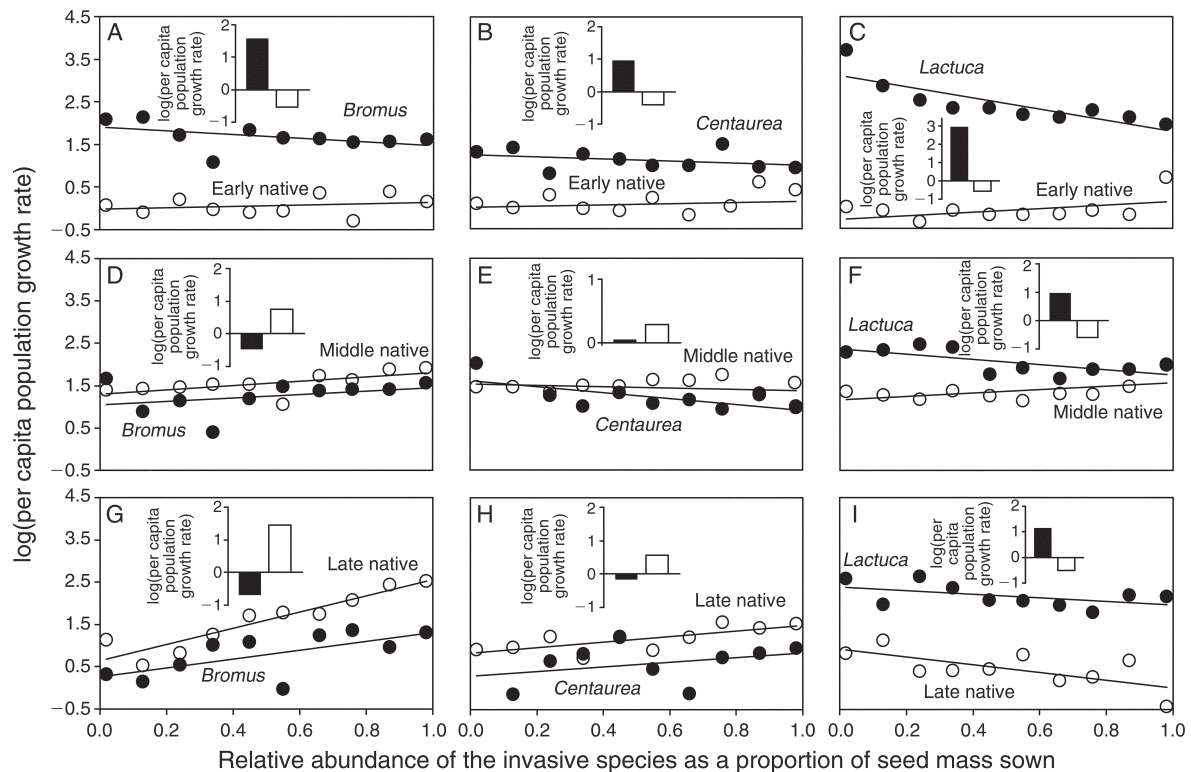


FIG. 3. Main panels show the per capita population growth rate of an invader and native community growing in competition with one another at different relative abundances. The species or community with the higher per capita growth rate at all relative abundances is the predicted winner. In panel E, the invader and native community each have a higher growth rate when they are at low relative abundance (their lines cross), yielding a prediction of coexistence. The insets show the theoretically predicted per capita growth rates when rare, based on the fitted parameters of Eqs. 1 and 2, for each pair of an invader and native community. When one has a positive growth rate when rare and the other a negative growth rate, competitive exclusion is the predicted outcome. If both have positive growth rates when rare, coexistence is the predicted outcome. Solid bars and symbols indicate the invader and open bars and symbols indicate the native community. Notice that the model predicted outcome in the insets matches the prediction from the relative abundance manipulation in nine of nine pairs.

These findings are among the most direct evidence that phenology differences between species allow invasions via the classic “vacant niche” mechanism (Yeaton et al. 1977, Veresoglou and Fitter 1984, Rathcke and Lacey 1985, Fargione and Tilman 2005, Hooper and Dukes 2010). In contrast to a resource preemption mechanism, phenological offsets of this sort should not favor the exotic over native, or early vs. late species, but instead favor whichever species is at low relative abundance. This is the invader when first introduced, but also native species driven to low relative abundance later in the invasion.

The second mechanism by which phenology differences favor invasion in our system involves later phenology giving species a fitness advantage. Invaders later than their native competitors produced up to 100 times more seeds per seed loss than their native counterparts (Fig. 2C), and this translated into a large fitness advantage (Fig. 2B). Surprisingly, this fitness advantage did not arise from greater resistance to competition by later taxa (Fig. 2D), despite their greater

biomass accumulation (Fig. 1). Prior work in California grasslands suggests that early taxa can suppress their later counterparts via nitrogen preemption early in the season (Mooney et al. 1986, Hooper and Dukes 2010), while later taxa suppress earlier competitors during shared periods of resource uptake, especially when water is limiting (Dukes 2001, 2002, Hulvey and Zavaleta 2012).

Regardless of the mechanism, the fitness advantage of later phenology was not unique to the invaders; later native communities were also favored over earlier invaders. This advantage led to our prediction that later native communities could sometimes exclude earlier invaders, demonstrating that phenological differences can sometimes facilitate invasion resistance. The consistency of the phenology–fitness relationships within both the three exotic taxa and the three native communities, each composed of three native species, is a key result. It suggests that the fitness advantage of a later phenology in our system is unlikely to reflect idiosyncrasies of the invaders selected for study, a potential concern when

logistical constraints limit our study to three invaders against three native communities.

In general, the fitness differences arising from later phenology overwhelmed the resulting niche differences, making coexistence more difficult between phenologically offset competitors. Indeed, the lone case of possible coexistence involved the invader and native community with intermediate phenology (Fig. 3E), a pair with little phenological offset. The success of later phenology invaders has been documented across a wide range of plant life forms and ecosystems including herbs, shrubs, and trees in the Mediterranean Basin (Celesti-Grapow et al. 2003, Lloret et al. 2005, Godoy et al. 2009b, Arianoutsou et al. 2010) and in central Europe (Knapp and Kühn 2012); lianas and shrubs in temperate forests (Fridley 2012) and riparian zones of North America (Resasco et al. 2007, Xu et al. 2007); and annuals in temperate and mediterranean grasslands (Benefield et al. 2001, Crone et al. 2009). Our finding that later phenology enhances invasion primarily through fitness advantages may apply to these systems as well.

Mechanisms underlying the stabilizing niche and relative fitness differences

The success of late-season invaders is commonly thought to rest on their ability to acquire unused resources left after the senescence of their competitors (Dyer and Rice 1999, Gerlach 2004). Indeed, Hooper and Dukes (2010) and Hulvey and Zavaleta (2012) showed that late annual plants in California grasslands best prevented the establishment of late invaders, and likely did so via competition for water (Dukes 2001, 2002). Our results suggest that invader use of summer resources not only translates into a stabilizing niche difference from earlier competitors, but to a greater extent, drives a fitness advantage, as the extra resources (evidenced by larger size), allow later taxa to produce far more offspring per seed loss than the earlier taxa (Fig. 2C). If so, the key influence of phenology on fitness differences may arise more from species differences in phenological breadth (the duration of growth, Wolkovich and Cleland 2010) than the timing of biomass accumulation per se, a hypothesis supported by prior studies (Gerlach and Rice 2003, Pyšek et al. 2003). In systems where the timing of growth is less strongly associated with phenological breadth (e.g., Fargione and Tilman 2005), phenology may be less related to fitness differences.

Water availability in California grasslands is spatially and temporally heterogeneous, raising interesting questions about how our results might change across years and locations. We have observed that many of our focal taxa exhibit earlier phenology in drier years. If a drier, compressed, growing season also means less phenological offset between species, this would potentially weaken the stabilizing niche difference between the resident species and the invader, and the relationship in Fig. 2A could potentially be used to predict this effect.

In addition, drier years or sites have lower summer water availability, a resource particularly important for late-season taxa. Thus the fitness advantage of late invaders would likely be less under drier conditions. Still, our prior work with the native annual species suggests that the fitness advantages of a late phenology persist in a dry year (Levine and HilleRisLambers 2009), and that the latest phenology species dominates across years (J. Levine, *unpublished data*).

Although our results generally predict the exclusion of earlier taxa (native or exotic) by later phenology competitors, we do observe species possessing very different phenologies co-occurring in the area surrounding our study system. This likely reflects differences between these taxa in how they respond to environmental heterogeneity occurring at spatial scales larger than our study plots but still within their dispersal range. More generally, the interaction between phenology and environmental variability in determining niche and fitness differences is a promising research avenue deserving further attention.

Our results also bear on understanding the success of early invaders in California such as the exotic annual grasses. Based on results here, we suggest that these species benefit from a stabilizing niche difference related to their phenological offset from later competitors. However, while a niche difference may help their establishment, it does not explain the dominance of these species in many locations in California. In fact, their dominance would not be predicted by our results suggesting fitness advantages for later taxa. Two factors likely explain this discrepancy. First, our experiments focused on forbs, with only two of 12 focal taxa being grasses. Second, our experiments and models cannot easily incorporate between-year interactions, as might occur with soil or litter feedbacks, processes that may contribute to the dominance of grasses, especially those with persistent thatch (Bartolome et al. 1980, Brandt et al. 2009).

Because we are using existing phenological variation among exotic and native taxa to evaluate the role of phenology in determining invasion success, our approach is ultimately correlational. And therefore, all factors correlated with phenology can contribute the outcomes observed. For example, later plants tend to root deeper in the soil and thus part of the stabilizing niche difference arising from phenological offset could reflect rooting depth differences. Other work in California grasslands has shown higher C:N ratios with later phenology (Gulmon et al. 1983); so fitness differences could also reflect physiological differences. We therefore view our results as reflecting the syndrome of traits associated with different phenologies in both native and exotic California annuals. Moreover, the consistency across native and exotic taxa in the relationships between phenology and niche differences, fitness differences, and biomass (Fig. 1 and Fig. 2A and B) suggest the existence of such a syndrome in this

system. More generally, we speculate that phenology is often associated with other traits in ecological communities, and that past results connecting phenology to invasion success likely reflect the correlated influence of multiple species traits.

Relevance for observational approaches

Much of our past understanding of how phenology relates to invasion success comes from surveys of invader and native phenology in different parts of the world (Godoy et al. 2009a, Wolkovich and Cleland 2010, Knapp and Kühn 2012), or detailed descriptions from individual sites (Fridley 2012). Even if we assume that the pool of introductions matches the native community in its phenological distribution (which is sometimes incorrect, see Godoy et al. [2009a]), and that successful invaders are those that compete well with the resident community, the simple outcome of competition does not fully reveal the mechanisms driving invasion success.

One could infer from the success of the latest invader and latest native community in the relative abundance manipulation (Fig. 3) that later phenology gave species increased fitness advantages (Fig. 2B). However, whether the dominance of later taxa relates to demographic advantages or greater competitive resistance could not be determined. Moreover, the role of phenology differences in driving stabilizing niche differences is entirely hidden in the competitive outcomes; coexistence did not increase with increasing phenological offset, and in fact, occurred only once in the experiment. We therefore conclude that fully understanding the role of phenology in determining invasion success requires parsing out the influences of phenology on niche and fitness differences.

In sum, understanding that phenology differences more strongly relate to fitness differences than to niche differences in our system suggests that invaders later than the resident community are more likely to eliminate than to coexist with native annual competitors. Understanding that phenology nonetheless underlies a stabilizing niche difference suggests that mowing, burning, or other management actions that minimize the fitness advantage of invaders might be particularly successful in promoting the persistence of phenologically offset native competitors. More generally, our findings show that both stabilizing niche differences and relative fitness advantages can combine to favor the invasion of species phenologically different from the residents of the communities they invade.

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SUPPLEMENTARY MATERIAL

Appendix A

Description of niche and fitness differences for our annual plant population model in a way that closely corresponds to how those terms are defined by Chesson (2000) for the Lotka-Volterra model (*Ecological Archives* E095-060-A1).

Appendix B

Mean and standard error values for the competitive coefficients and vital rates calculated to parameterize our annual plant population model (*Ecological Archives* E095-060-A2).